

Lab 3: GENE EXPRESSION

You are being given some samples of bacteria that contain protein coding sequences under the regulation of different bacterial promoters.

You have 6 different samples.

- *E. coli* containing a plasmid encoding the Maltose Binding Protein (MBP) under the regulation of a bacterial promoter.
- *E. coli* containing a plasmid encoding the Green Fluorescent Protein (GFP) under the regulation of a bacterial promoter.
- *E. coli* containing a plasmid encoding the Red Fluorescent Protein (RFP) under the regulation of a bacterial promoter.

All three of these were incubated with or without Isopropyl thio-galactosidase (IPTG), an analog of lactose that binds to the Lac repressor.

Look at the attached sheets for information about these.

You have two different kinds of acrylamide gels.

- Denaturing polyacrylamide gels that contain SDS. These will break almost all inter and intra-protein non-covalent bonds.
- Native polyacrylamide gels that do not contain SDS. These will not break inter and intra-protein non-covalent bonds. The proteins will be left in their native conformation.

Questions: 1. Which of the bacterial promoters are inducible by IPTG?
2. Do the fluorescent proteins form dimers or other multimers?

The experimental procedure is listed below. However, we have done the bacterial growth and induction steps for you, and we have prepared the acrylamide gels for you. Parts of the procedure that you are not responsible for are shaded.

1. Native protein preparation

Liquid bacterial culture

- 1.1. Add 2 ml of LB media with 50 ug/ml ampicillin to a 50 ml tube
- 1.2. Sterilize bacterial loop in flame and cool
- 1.3. With the loop, carefully remove one bacterial colony from the agar plate and transfer it to the 5 ml of LB media
- 1.4. Incubate overnight at 37° C with shaking

Prepare exponentially growing bacteria

- 1.5. Add 2 ml of LB media with 50 ug/ml ampicillin to a 50 ml tube
- 1.6. Inoculate the media with 200 ul of the overnight bacterial culture
- 1.7. Incubate for one hour at 37° C with shaking

Induction of the modified lac promoter

- 1.8. Add 10 ul of 100 mM IPTG to the culture
- 1.9. Incubate for two hours at 37° C with shaking

Lyse bacteria with lysozyme and detergent

- 1.10. Transfer 1.5 ml of the bacterial culture to a microfuge tube
- 1.11. Centrifuge for 1 min to pellet the cells
- 1.12. Remove and discard the supernatant
- 1.13. Following steps must be done on ice
- 1.14. Suspend the pellet in 100 ul of Protein Extraction Buffer (PEB)

1.15. Add 2 ul of 10 mM PhenylMethylSulfonyl Fluoride to inhibit protease activity

HAZARD: PhenylMethylSulfonyl Fluoride (PMSF) is extremely toxic.

1.16. Add 4 ul of 20 mg/ml lysozyme to degrade the bacterial cell wall

1.17. Add 1 ul of 1mg/ml DNase to degrade bacterial DNA

1.18. Incubate on ice with occasional shaking for 10 minutes

1.19. Add 10 ul of 12mg/ml deoxycholic acid to lyse bacterial membrane

1.20. Incubate 10 min on ice

1.21. Incubate 10 min at 37° C

Remove insoluble material (eg cell wall components) by centrifugation

1.22. Centrifuge for 5 min at 4° C

1.23. Remove 30 ul of supernatant to a new tube

1.24. Add 30 ul of 2X PAGE Sample Buffer

1.25. Mix and store on ice

2. Polyacrylamide Gel Electrophoresis (PAGE)

Prepare separation gel

2.1. Prepare sandwich according to manufacturers instructions

2.2. Put in sample loading comb and mark (on the outside glass) a position 1.5 cm below the bottom of the comb

2.3. Prepare suitable volume of SDS PAGE separating gel.

Hazard: Unpolymerized acrylamide is a potent neurotoxin (found in potato chips).

2.4. Gently mix sample while degassing under vacuum for 10 min

2.5. Add TEMED to catalyze polymerization

2.6. Mix gently

2.7. Pour up to the top of the outer plate

2.8. Wet the comb with the gel and gently insert it into the gel - start from one end first.

2.9. Wait one hour for polymerization.

2.10. Remove well forming comb

2.11. Immediately fill upper running buffer tank with SDS PAGE running buffer (above the level of the inner glass plate) and rinse wells to remove any unpolymerized gel

Electrophorese samples

2.12. Remove 10 ul of sample using a fine tip micropipette

2.13. Add the sample to the bottom of the gel well

2.14. Put on the top of the electrophoresis unit and turn on power supply.

2.15. Electrophorese until bromophenol blue marker is near the bottom of the gel

HAZARD: high voltage

2.16. Remove glass plate from PAGE gel

2.17. Cut off and discard stacking gel

3. Protein detection

Fluorescence

3.18. Place gel on UV light box. Replace UV shield

3.19. Illuminate with long wave UV

3.20. Photograph gel.

Coomassie blue staining

3.21. Put separating gel into a glass dish. Add 100 ml of Coomassie blue stain solution

3.22. Incubate at room temperature for 20 min with shaking

3.23. Remove and save stain solution

3.24. Add 200 ml of Coomassie destain solution (if destaining is needed)

3.25. Incubate at room temperature for 20 min with shaking

3.26. Remove and discard destain

3.27. Repeat destaining until gel is clear